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(54) **SHIELDING PARTICLES COATED WITH ELECTRICAL INSULATION**

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(57)

ABSTRACT

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(60) Provisional application No. 63/563,243, filed on Mar. 8, 2024.

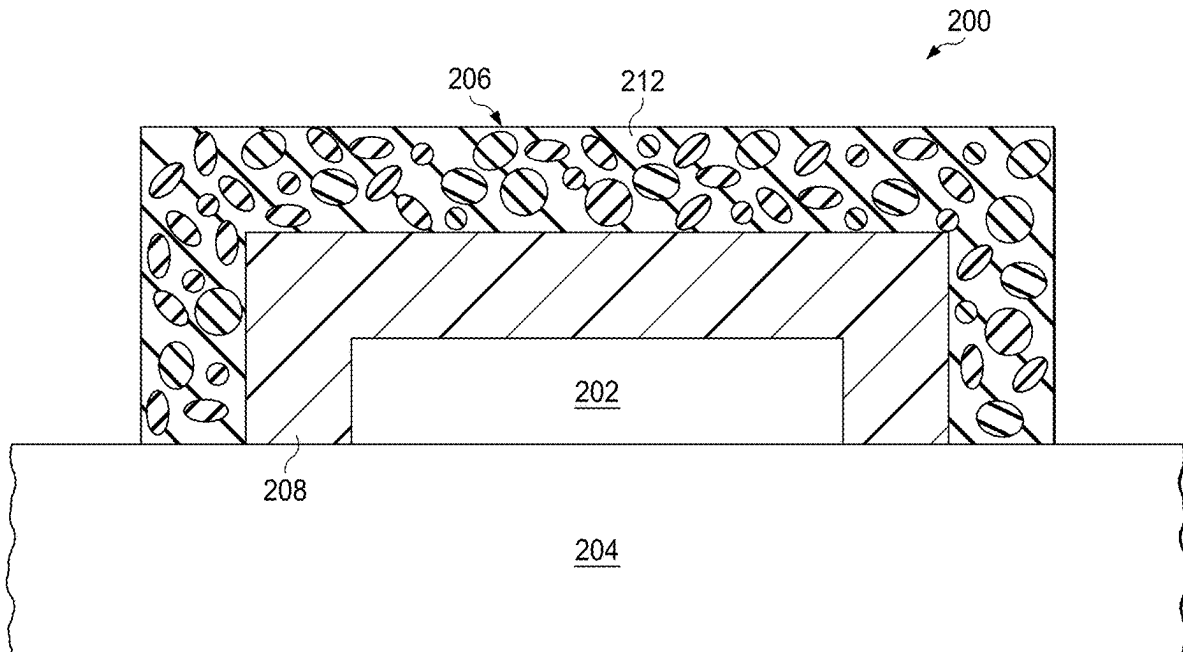
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Methods to coat shielding particles with an electrically insulating coating, disperse the coated shielding particles in an base material to form a mold structure; and position the mold structure proximate a die of an integrated circuit package to shield the die from radiation. Devices comprising: a die; and a mold structure proximate the die, the mold structure comprising: an base material; and shielding particles comprising an electrically insulating coating, wherein the shielding particles are dispersed in the base material.



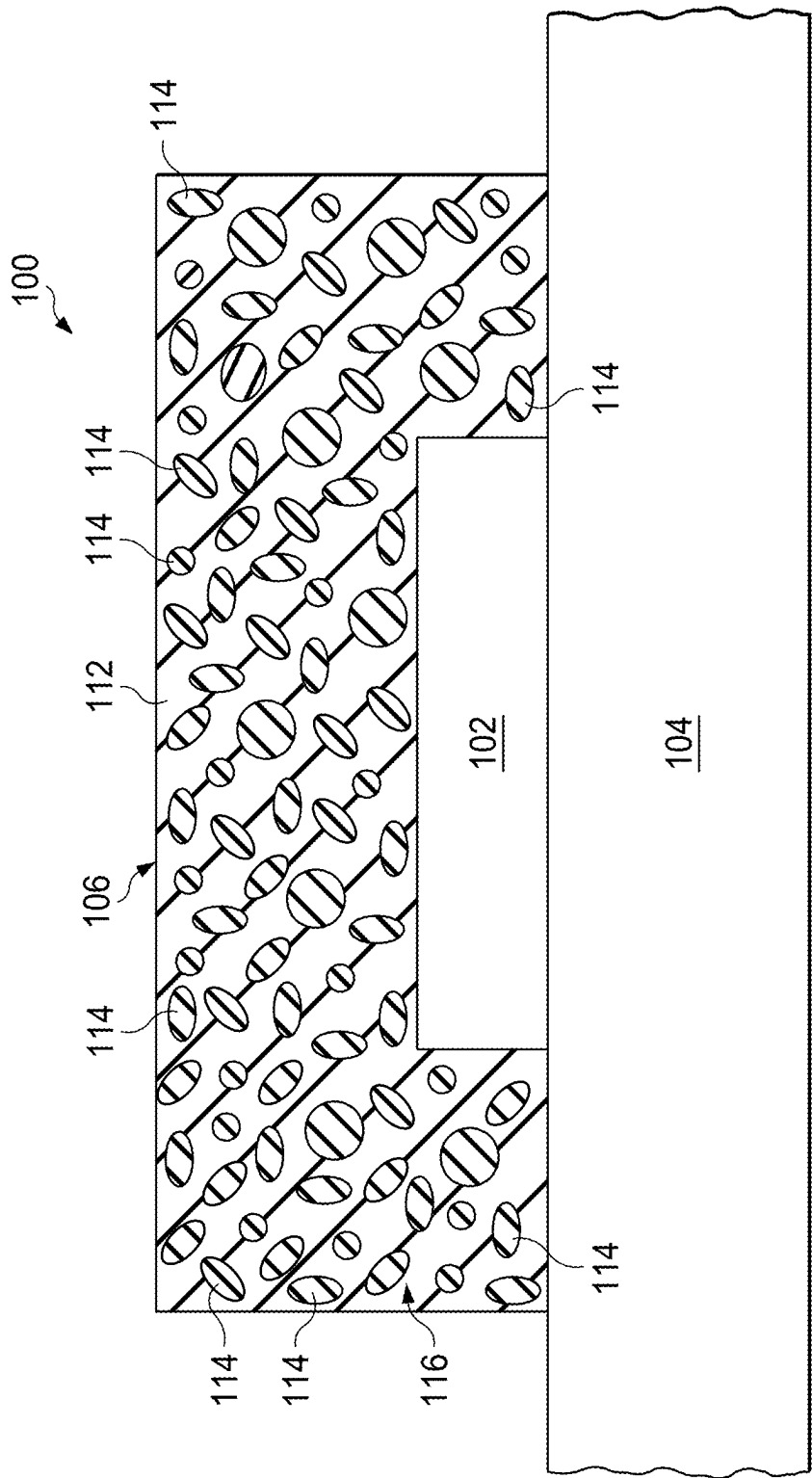


FIG. 1

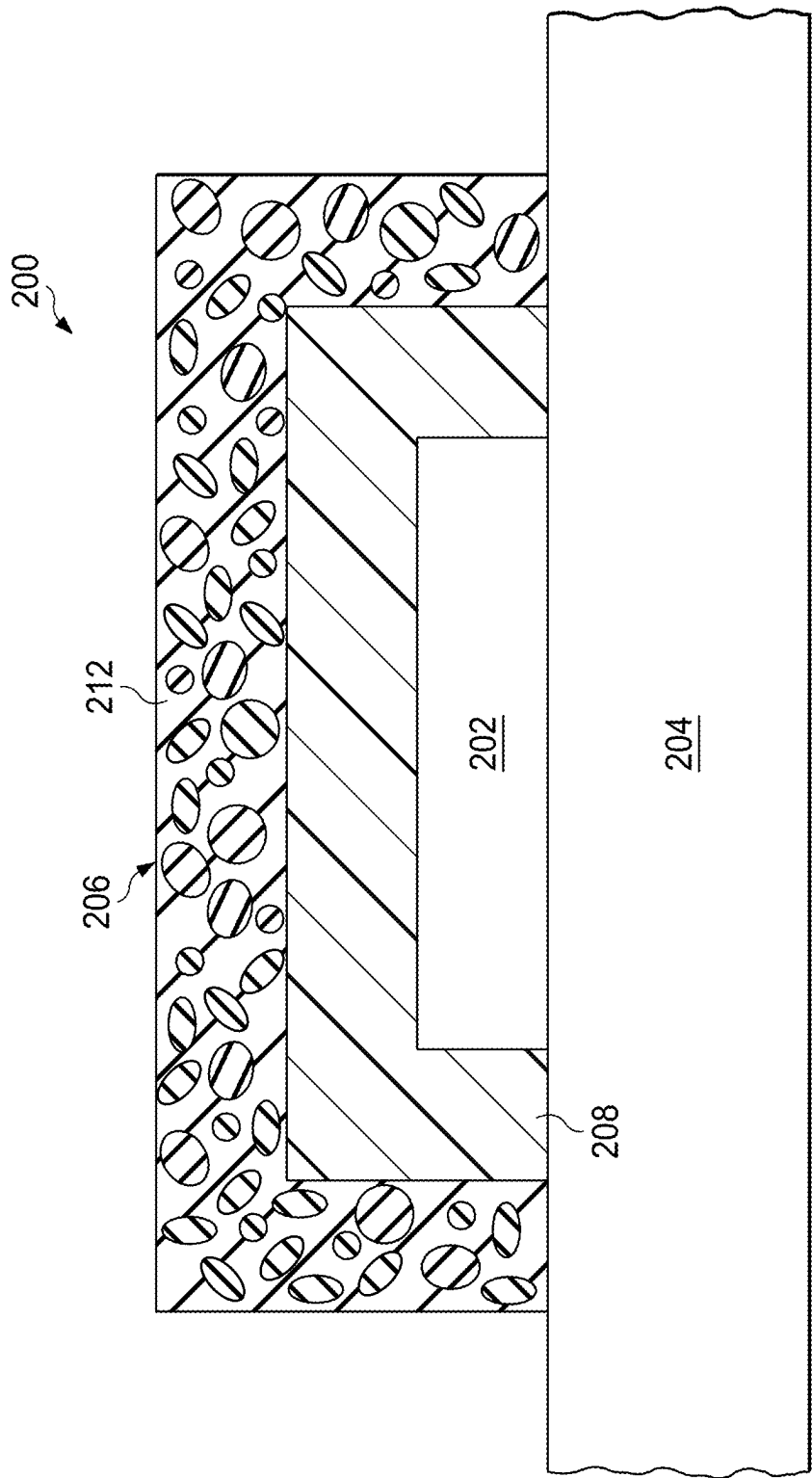


FIG. 2

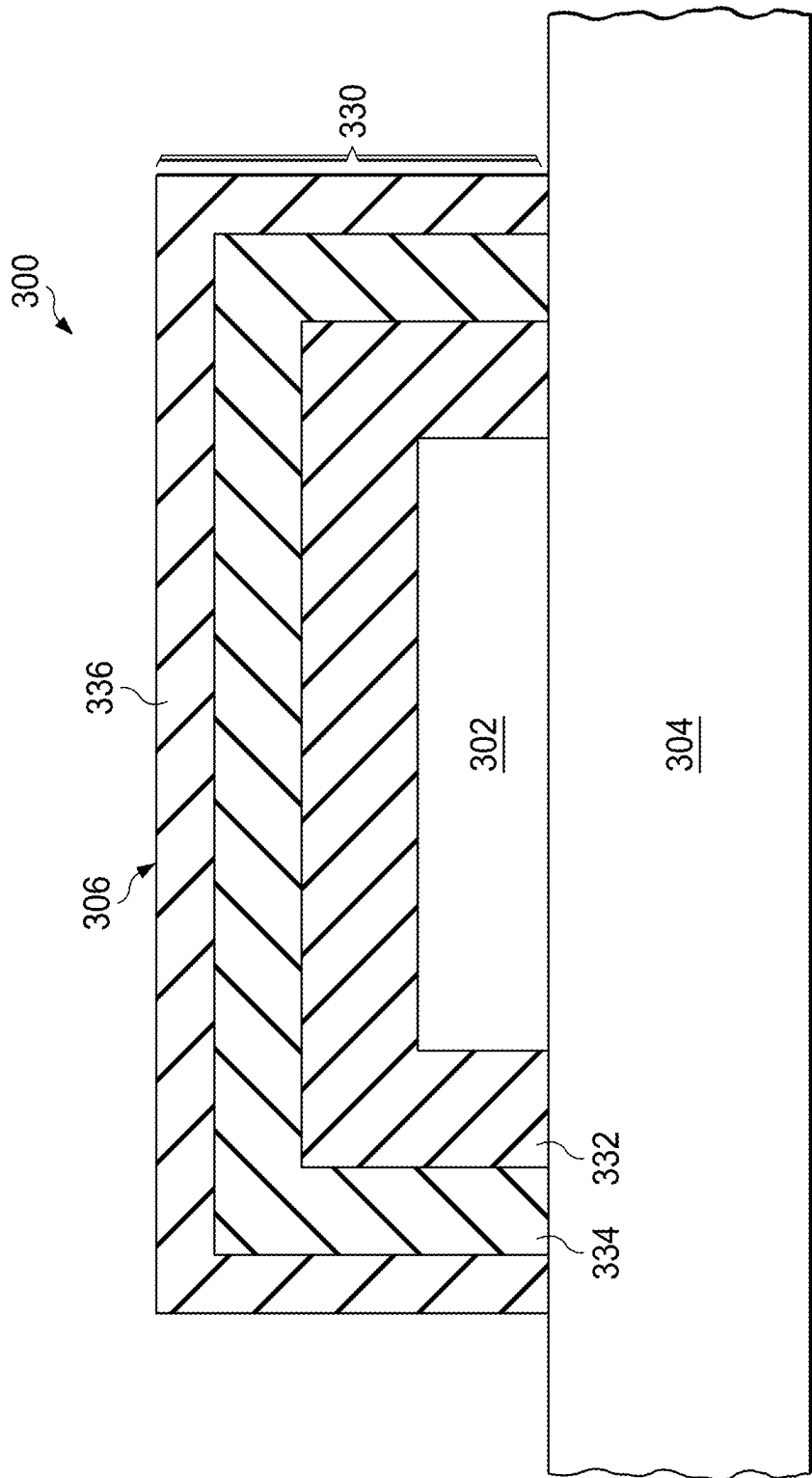


FIG. 3

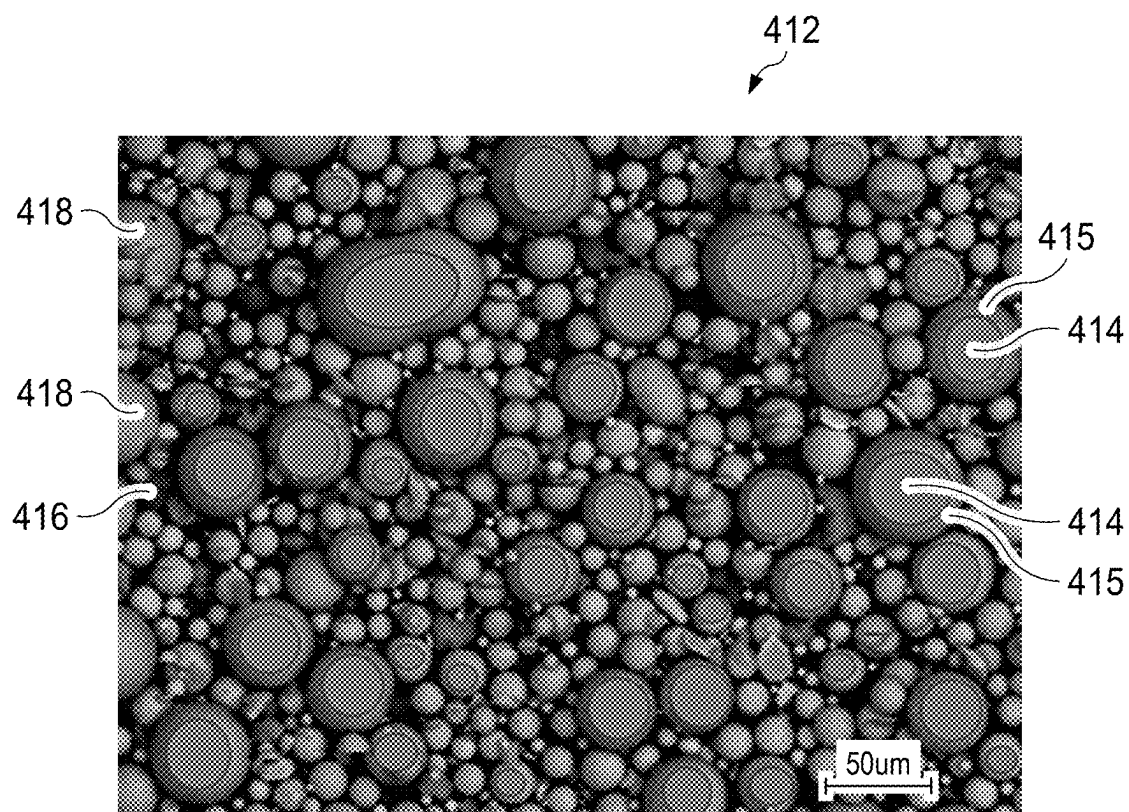
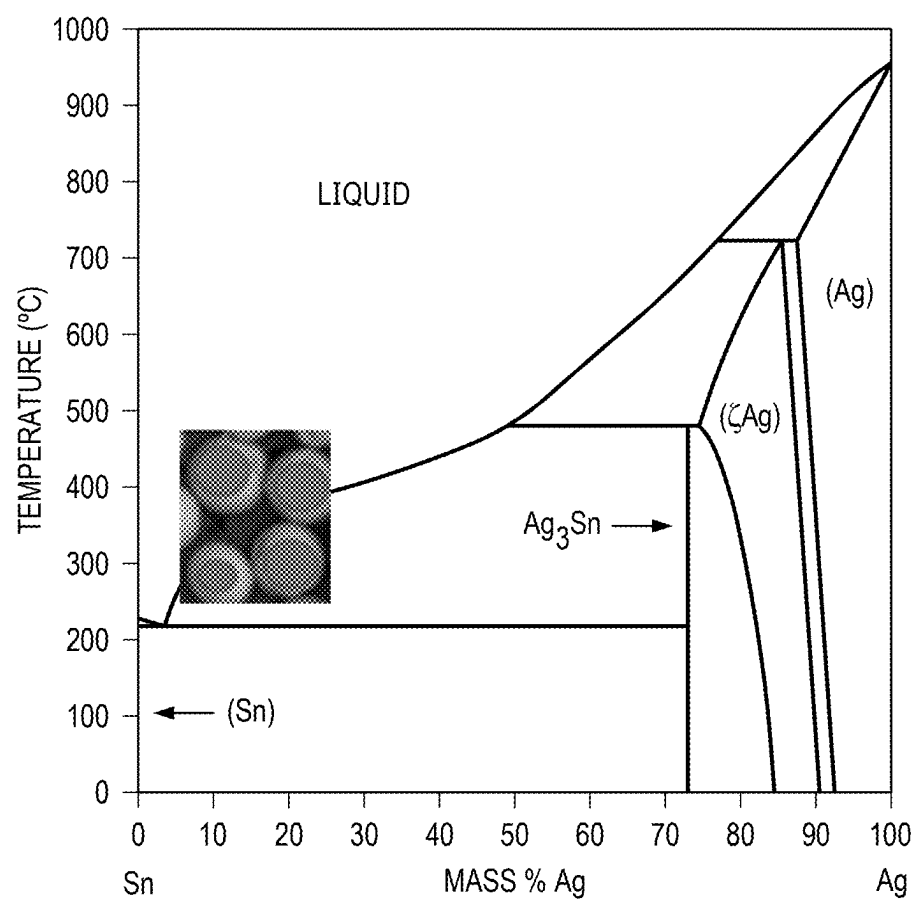
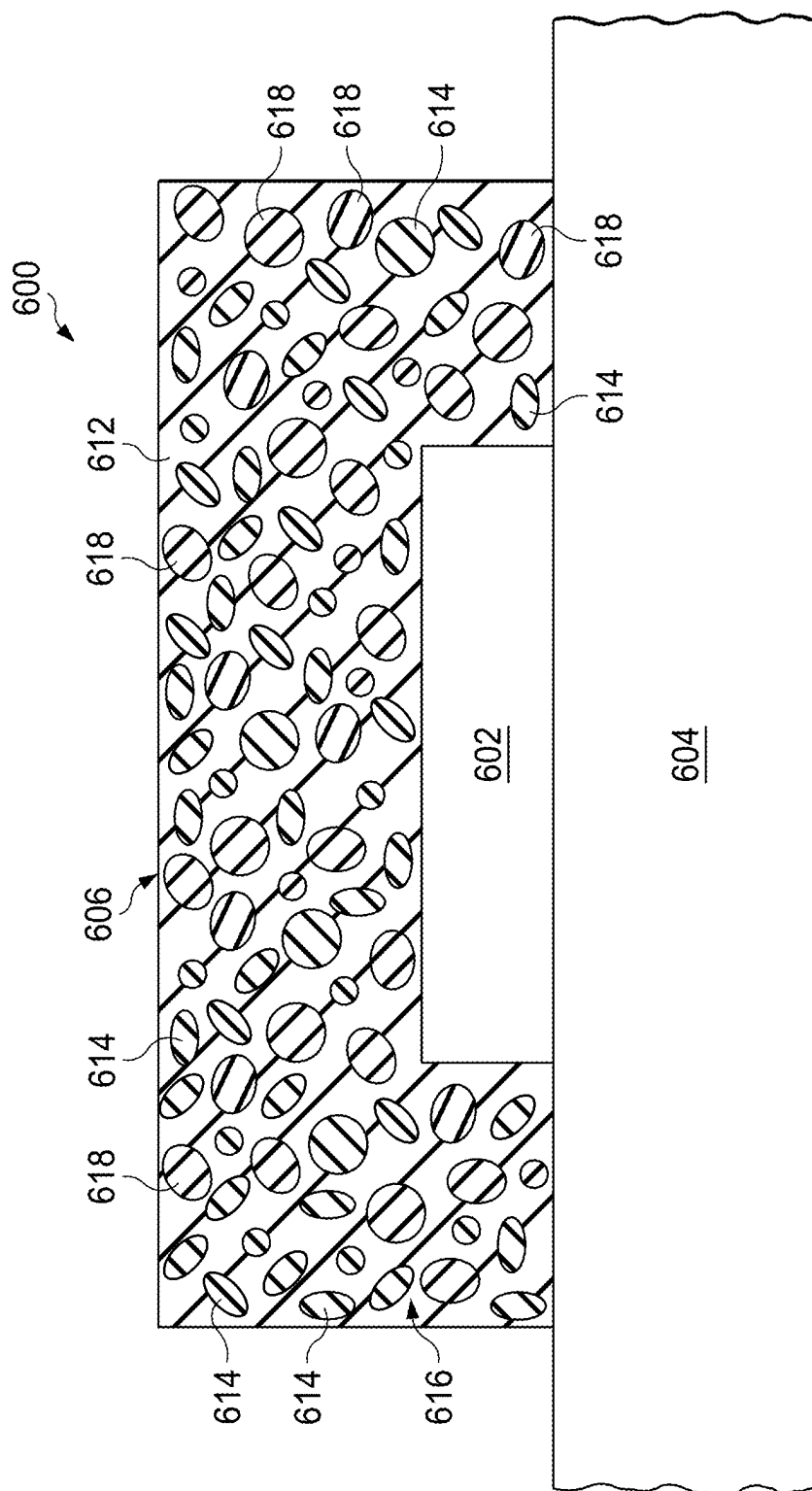


FIG. 4

FIG. 5





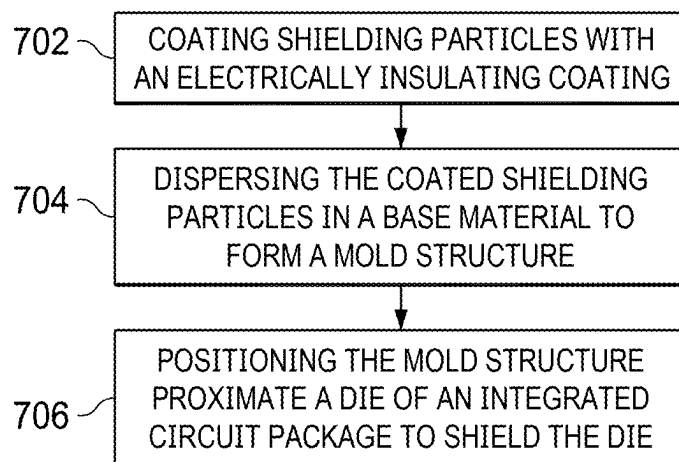


FIG. 7

SHIELDING PARTICLES COATED WITH ELECTRICAL INSULATION

PRIORITY STATEMENT

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/563,243, filed Mar. 8, 2024, the contents of which are hereby incorporated in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to radiation shielded die packages, in particular, die packages comprising a shielding compound having shielding particles coated with an electrical insulation coating and dispersed in a base material.

BACKGROUND

[0003] There is a growing need for Radiation Tolerant (RT) semiconductors for applications like avionics as well as Low Earth Orbit (LEO) satellites. Normally to create these types of products, either a specific expensive manufacturing fabrication process is used, or the products are placed in expensive ceramic packages.

[0004] There is a need for RT semiconductors with reduced cost.

SUMMARY

[0005] Most semiconductors are over-molded with a compound comprising an epoxy base material and over 80% silicon dioxide (SiO_2) filler particles for structural stability. An aspect replaces, at least in part, these SiO_2 filler particles with a radiation blocking material (such as an Ag—Sn alloy) coated with SiO_2 , to achieve a level of radiation blocking without needing to use a ceramic package or alter the fabrication process.

[0006] Aspects provide a method comprising: coating shielding particles with an electrically insulating coating; dispersing the coated shielding particles in a base material to form a mold structure; and positioning the mold structure proximate a die of an integrated circuit package to shield the die from radiation.

[0007] According to an aspect, this is provided a method as described above, wherein the shielding particles comprise at least one of boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN), tungsten nitride (W_3N_2), tin oxide (SnO_2), copper (I) oxide (Cu_2O), or copper (II) oxide (CuO).

[0008] According to an aspect, this is provided a method as described above, wherein the shielding particles comprise a material from at least one of the Cobalt oxide family, the Nickel oxide family, the Neodymium oxide family, and the Iron oxide family.

[0009] According to an aspect, this is provided a method as described above, wherein the electrically insulating coating comprises silicon dioxide (SiO_2).

[0010] According to an aspect, this is provided a method as described above, wherein the base material comprises polymer, silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber, isoprene, resin, or epoxy.

[0011] According to an aspect, this is provided a method as described above, comprising dispersing a plurality of silicon dioxide (SiO_2) filler particles in the base material to form the mold structure.

[0012] According to an aspect, this is provided a method as described above, wherein dispersed particles comprise 50% to 95% shielding particles and 50% to 5% silicon dioxide filler particles.

[0013] According to an aspect, this is provided a method as described above, wherein positioning the mold structure proximate the die comprises the mold structure at least partially encapsulating the die.

[0014] According to an aspect, this is provided a method as described above, wherein positioning the mold structure proximate the die comprises comprising positioning first and second shielding layers, wherein the first and second shielding layers comprise different concentrations of the coated shielding particles.

[0015] According to an aspect, there is provided a device comprising: a die; and a mold structure proximate the die, the mold structure comprising: a base material; and shielding particles comprising an electrically insulating coating, wherein the shielding are dispersed in the base material.

[0016] An aspect provides a device as described above, wherein the shielding particles comprise at least one of boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN), tungsten nitride (W_3N_2), tin oxide (SnO_2), copper (I) oxide (Cu_2O), or copper (II) oxide (CuO).

[0017] An aspect provides a device as described above, wherein the shielding particles comprise a material from at least one of the Cobalt oxide family, the Nickel oxide family, the Neodymium oxide family, and the Iron oxide family.

[0018] An aspect provides a device as described above, wherein the electrically insulating coating comprises silicon dioxide (SiO_2).

[0019] An aspect provides a device as described above, wherein the base material comprises a material selected from polymer, silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber, isoprene, resin, and epoxy.

[0020] An aspect provides a device as described above, wherein the mold structure comprises silicon dioxide (SiO_2) filler particles.

[0021] An aspect provides a device as described above, wherein particles in the mold structure comprise 50% to 95% shielding particles and 50% to 5% silicon dioxide (SiO_2) filler particles.

[0022] An aspect provides a device as described above, comprising an encapsulate at least partially encapsulating the die, wherein the mold structure is proximate the encapsulate.

[0023] An aspect provides a device as described above, wherein the mold structure comprises first and second shielding layers, wherein the first and second shielding layers comprise different concentrations of shielding particles.

[0024] According to an aspect, there is provided an integrated circuit package comprising: a die carrier; a die mounted on the die carrier; and a mold structure proximate the die, the mold structure comprising: a base material; shielding particles comprising an electrically insulating coating, the shielding particles dispersed in the base material; and silicon dioxide (SiO_2) filler particles dispersed in the base material.

[0025] An aspect provides a system as described above, wherein the shielding particles comprise at least one of boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN), tungsten nitride (W_3N_2), tin oxide

(SnO_2), copper (I) oxide (Cu_2O), or copper (II) oxide (CuO), wherein the electrically insulating coating comprises silicon dioxide (SiO_2), wherein the base material comprises polymer, silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber, isoprene, resin, or epoxy, and wherein particles in the mold structure comprises 50% to 95% shielding particles and 50% to 5% silicon dioxide (SiO_2) filler particles.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The figures illustrate examples of semiconductors that are over-molded with epoxy a radiation blocking material (such as an Ag—Sn alloy) coated with SiO_2 , to achieve a level of radiation blocking without needing to use a ceramic package or alter the fabrication process.

[0027] FIG. 1 is a cross-sectional side view showing an example integrated circuit package including a die mounted on a die carrier, and a mold structure (e.g., a mold encapsulation) at least partially encapsulating the mounted die.

[0028] FIG. 2 shows a cross-sectional side view of an example integrated circuit package having a die on a die carrier and having an encapsulate that encapsulates the die, and a mold structure that covers the encapsulate to shield the die from radiation.

[0029] FIG. 3 shows a cross-sectional side view of an example integrated circuit package having a die on a die carrier and having a mold structure encapsulating the die, wherein the mold structure comprises layers.

[0030] FIG. 4 is an SEM image of a shielding compound with shielding particles coated with an electrical insulator (such as SiO_2), wherein the shielding particles are sphere and colloid particles of various sizes.

[0031] FIG. 5 is a phase diagram showing alloys like Ag—Sn.

[0032] FIG. 6 is a cross-sectional side view showing an integrated circuit package including a die mounted on a die carrier, and a mold structure at least partially encapsulating the mounted die, wherein the mold structure has a shielding compound with shielding particles coated with an electrical insulator and filler particles dispersed in a base material.

[0033] FIG. 7 is a flow chart of a method of manufacturing a die package with a shielding mold structure.

[0034] The reference number for any illustrated element that appears in multiple different figures has the same meaning across the multiple figures, and the mention or discussion herein of any illustrated element in the context of any particular figure also applies to each other figure, if any, in which that same illustrated element is shown.

DESCRIPTION

[0035] According to an aspect, there is provided a shielding compound to encapsulate dies in packages. The shielding compound includes shielding particles, which are coated to make them electrically nonconductive. The coated shielding particles are dispersed in a base material, for example a polymer such as an epoxy resin.

[0036] FIG. 1 is a cross-sectional side view showing an example integrated circuit package **100** including a die **102** mounted on a die carrier **104**, and a mold structure **106** (e.g., a mold encapsulation) at least partially encapsulating the mounted die **102**. The mold structure **106** may be proximate the die **102**.

[0037] The die **102** may comprise any type of die, chip (e.g., silicon substrate having an integrated circuit formed thereon), or other integrated circuit device (e.g., including analog devices, digital devices, or a mixture of analog and digital devices) that generates or outputs heat. For example, the die **102** may comprise a microprocessor (e.g., a central processing unit (CPU) chip), a microcontroller (MCU), an application specific IC (ASIC), a graphics processing unit (GPU), a digital signal processor (DSP), an A/D converter or D/A converter, or memory (e.g., Flash memory, random access memory (RAM), read only memory (ROM), e.g., electrically erasable programmable read-only memory (EEPROM), or other memory), or a system-on-chip (SoC) device.

[0038] The die carrier **104** may comprise any structure on which the die **102** may be mounted, for example a printed circuit board (PCB), a lead frame, an interposer, a heat sink, or another die. The die **102** may be mounted on the die carrier **104** in any suitable manner, for example solder mounting, adhesive bonding (e.g., using an epoxy), flip-chip bonding, or eutectic bonding.

[0039] As shown in FIG. 1, in some examples mold structure **106** may comprise a shielding compound **112** includes shielding particles **114** dispersed in, or otherwise combined, in a base material **116**, which base material **116** may be, for example, a polymer such as an epoxy resin or a polymer resin. In some examples, the shielding compound **112** includes a mixture of shielding particles **114**, filler particles comprising silicon dioxide (SiO_2) particles, and a base material **116** comprising an epoxy resin. The mold structure **106** may be proximate the die **102** to shield the die from radiation.

[0040] As used herein, a “compound” may refer to one element or substance, or a mixture or other combination of multiple elements or substances. The term particles, as used herein, refers to a particle having a maximum dimension between 1 nanometer and 1000 micrometers, and may be spherical, or colloidal shaped, without limitation. Particles may have a maximum dimension between 1 and 300 micrometers, or 50-80 micrometers. Shielding particles and filler particles are described below.

[0041] In some examples the shielding particles **114** comprise at least one of gold-tin, silver-tin, tungsten, antimony, bismuth, and any heavy metal, without limitation. Heavy metals include metal with high density (for example 5 g/cm^3), high atomic weights (for example greater than 63.5 gmol^{-1}), or atomic numbers greater than 20, such as for example, boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN), tungsten nitride (W_3N_2), tin oxide (SnO_2), copper (I) oxide (Cu_2O) (i.e., cuprous oxide), or copper (II) oxide (CuO) (i.e., cupric oxide), antimony (Sb), tin (Sn), tungsten (W) particles, without limitation, wherein the heavy metal particles may be coated or encapsulated with an electrically nonconductive material, such as silicon dioxide (SiO_2). The shielding particles **114** are coated or encapsulated with an electrically nonconductive material, such as silicon dioxide (SiO_2) (also called silica). The shielding particles **114** may be coated with an electrically nonconductive material to reduce the possibility of an electrical conduction path that could short pins of an integrated circuit package together.

[0042] In some examples, the shielding particles **114** are dispersed in, or otherwise combined with, the base material **116**. The base material **116** may comprise, for example, an

elastomer (e.g., silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber or isoprene), a thermoset (e.g., thermoset resin), or other molding compound, which may be supplied in the form of pellets, liquids, or powders, for example. In some examples the shielding particles **114** may shield the die **102** from ionizing radiation, magnetic fields, or a combination of ionizing radiation and magnetic fields. Shielding particles **114** to shield from magnetic fields may comprise material from the Cobalt oxide family, the Nickel oxide family, the Neodymium oxide family and the Iron oxide family. The shielding particles **114** may comprise a material from at least one of the Cobalt oxide family, the Nickel oxide family, the Neodymium oxide family, and the Iron oxide family. Shielding particles **114** to shield from ionizing radiation may include mu-metal or hematite (Fe_2O_3) particles, for example. Thus, shielding particles **114** need not be uniform, and may comprise a plurality of different types of shielding particles.

[0043] The shielding particles **114** may be coated or encapsulated with an electrically nonconductive material, such as silicon dioxide (SiO_2), by a SOL-GEL process or a spin-on-glass process.

[0044] The coated or encapsulated shielding particles **114** may be dispersed in or otherwise combined with a base material **116** to produce the shielding compound **112**. In some examples a surfactant **113** may (optionally) be added to enhance or expedite the dispersing of the shielding particles **114** in the base material **116**. The shielding particles **114** (with or without surfactant **113**) may be mixed or combined with the base material **116** in any suitable manner, e.g., using an agitation or ultrasonic vibration process.

[0045] FIG. 2 shows a cross-sectional side view of a package **200** having a die **202** on a die carrier **204**. An encapsulate **208** encapsulates the die **202** and comprises a material that may not shield the die **202** from radiation. A mold structure **206**, comprising shielding compound **212** covers the encapsulate **208** to shield the die **202** from radiation. The illustrated example shows shielding compound **212** formed over encapsulate **208**. Shielding compound **212** is similar in all respects to shielding compound **112** described above. The mold structure **206** may be proximate the die **202** to shield the die **202** from radiation.

[0046] FIG. 3 shows a cross-sectional side view of a package **300** having a die **302** on a die carrier **304** with a mold structure **306** comprising three shielding layers **330**. Other examples, not shown, include a single shielding layer, two shielding layers, or three, or more, shielding layers. Respective shielding layer(s) may include shielding particles dispersed in or otherwise combined with a base material. For example, respective shielding layer(s) may include shielding particles to shield the die **302** from ionizing radiation, magnetic fields, or a combination of ionizing radiation and magnetic fields, e.g., as discussed above regarding shielding compound **112**, with reference to FIG. 1. The mold structure **306** may be proximate the die **302** to shield the die **302**.

[0047] In examples with multiple shielding layers, different shielding layers **330** may include different types or different concentrations of shielding particles. For example, referring to the example shown in FIG. 3, shielding layer **332** may comprise magnetic shielding particles (e.g., to shield die **302** from magnetic fields) and shielding layer **334** may comprise shielding particles (e.g., to shield die **302** from ionic radiation). As another example, shielding layers

332, **334**, and **336** may comprise the same type of particles (e.g., magnetic shielding particles or shielding particles) but with a respective different concentration of particles, to thereby define a shielding gradient along a direction toward or away from the die **302**. For example, the shielding layers **332**, **334**, and **336** may provide an increasing degree of shielding in a direction toward the die **302** (e.g., wherein the shielding layer **332**, which may be termed inner shielding layer **332**, may provide a greater degree of shielding than shielding layer **336**, which may be termed outer shielding layer **336**). While three shielding layers **330** are shown in FIG. 3, any number of shielding layers may be used, without limitation, and individual layers may have any ratio of shielding particles to filler particles, without limitation. The shielding particles of the respective ones of shielding layers **330**, may be coated or encapsulated with an electrically nonconductive material, such as silicon dioxide (SiO_2).

[0048] FIG. 4 is an scanning electron microscope (SEM) of a shielding compound **412** with shielding particles **414** dispersed in a base material **416**. The shielding particles **414** are shown in cross-section to show that they are coated with an electrical insulator **415** (such as silicon dioxide (SiO_2)), wherein the shielding particles **414** are sphere and colloid particles of various sizes. Filler particles **418** may also be dispersed in the shielding compound **412**. The filler particles **418** may thicken or otherwise enhance the structural integrity of the base material **416**.

[0049] FIG. 5 is a phase diagram showing alloys like Ag—Sn allow for dialing in the radiation blocking ability while being able to hold up to processing temperatures and the mission profile of the end device. Sn has more effective radiation blocking properties than Ag. However, Sn has a relatively low melting point and may not hold up to processing temperatures by itself. By increasing the % of Ag in the alloy with Sn, the melting point increases. In some aspects, a high percentage of Sn compared to Ag may be used so the alloy will effectively block radiation without melting during package processing.

[0050] FIG. 6 is a cross-sectional side view showing an example integrated circuit package **600** including a die **602** mounted on a die carrier **604**, and a mold structure **606** (e.g., a mold encapsulation) at least partially encapsulating the mounted die **602**. As shown in FIG. 6, the mold structure **606** comprises a shielding compound **612**, which includes shielding particles **614** dispersed in a base material **616**, for example a polymer such as an epoxy resin. The shielding compound **612** also includes filler particles **618** (e.g., in the form of fumed silicon dioxide (SiO_2) or colloidal silicon dioxide (SiO_2)) to thicken or otherwise enhance the structural integrity of the base material **616** (e.g., epoxy). Thus, the shielding compound **612** may include shielding particles **614** and filler particles **618** dispersed in, or otherwise combined with, base material **616**. In one example, the particles may comprise 90%-95% shielding particles **614** and 10%-5% filler particles in the shielding compound **612**. In another example, particles may comprise 50%-95% shielding particles **114** and 50%-5% filler particles **618** in the shielding compound **612**.

[0051] FIG. 7 shows a flow chart of a method. Shielding particles are coated **702** with an electrically insulating coating. The coated shielding particles are dispersed **704** in a base material to form a mold structure. The mold structure is positioned **706** proximate a die of an integrated circuit package to shield the die from ionic radiation.

[0052] Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these disclosed examples.

1. A method comprising:
 - coating shielding particles with an electrically insulating coating;
 - dispersing the coated shielding particles in a base material to form a mold structure; and
 - positioning the mold structure proximate a die of an integrated circuit package to shield the die.
2. The method as in claim 1, wherein the shielding particles comprise at least one of boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN), tungsten nitride (W_3N_2), tin oxide (SnO_2), copper (I) oxide (Cu_2O), or copper (II) oxide (CuO).
3. The method as in claim 1, wherein the shielding particles comprise a material from at least one of the Cobalt oxide family, the Nickle oxide family, the Neodymium oxide family, and the Iron oxide family.
4. The method as in claim 1, wherein the electrically insulating coating comprises silicon dioxide (SiO_2).
5. The method as in claim 1, wherein the base material comprises polymer, silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber, isoprene, resin, or epoxy.
6. The method as in claim 1, comprising dispersing a plurality of silicon dioxide (SiO_2) filler particles in the base material to form the mold structure.
7. The method as in claim 6, wherein dispersed particles comprise 50% to 95% shielding particles and 50% to 5% silicon dioxide filler particles.
8. The method as in claim 1, wherein positioning the mold structure proximate the die comprises the mold structure at least partially encapsulating the die.
9. The method as in claim 1, wherein positioning the mold structure proximate the die comprises comprising positioning first and second shielding layers, wherein the first and second shielding layers comprise different concentrations of the coated shielding particles.
10. A device comprising:
 - a die; and
 - a mold structure proximate the die, the mold structure comprising:
 - a base material; and
 - shielding particles comprising an electrically insulating coating, wherein the shielding particles are dispersed in the base material.
11. The device as in claim 10, wherein the shielding particles comprise at least one of boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN),

tungsten nitride (W_3N_2), tin oxide (SnO_2), copper (I) oxide (Cu_2O), or copper (II) oxide (CuO).

12. The device as in claim 10, wherein the shielding particles comprise a material from at least one of the Cobalt oxide family, the Nickle oxide family, the Neodymium oxide family, and the Iron oxide family.

13. The device as in claim 10, wherein the electrically insulating coating comprises silicon dioxide (SiO_2).

14. The device as in claim 10, wherein the base material comprises a material selected from polymer, silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber, isoprene, resin, and epoxy.

15. The device as in claim 10, wherein the mold structure comprises silicon dioxide (SiO_2) filler particles.

16. The device as in claim 15, wherein particles in the mold structure comprise 50% to 95% shielding particles and 50% to 5% silicon dioxide (SiO_2) filler particles.

17. The device as in claim 10, comprising an encapsulate at least partially encapsulating the die, wherein the mold structure is proximate the encapsulate.

18. The device as in claim 10, wherein the mold structure comprises first and second shielding layers, wherein the first and second shielding layers comprise different concentrations of shielding particles.

19. An system comprising:

- a die carrier;
- a die mounted on the die carrier; and
- a mold structure proximate the die, the mold structure comprising:
 - a base material;
 - shielding particles comprising an electrically insulating coating, the shielding particles dispersed in the base material; and
 - silicon dioxide (SiO_2) filler particles dispersed in the base material.

20. The system as in claim 19,

wherein the shielding particles comprise at least one of boron nitride (BN), bismuth (Bi), bismuth oxide (Bi_2O_3), tantalum nitride (TaN), tungsten nitride (W_3N_2), tin oxide (SnO_2), copper (I) oxide (Cu_2O), or copper (II) oxide (CuO),

wherein the electrically insulating coating comprises silicon dioxide (SiO_2),

wherein the base material comprises polymer, silicone, polyurethane, chloroprene, butyl, polybutadiene, neoprene, natural rubber, isoprene, resin, or epoxy, and

wherein particles in the mold structure comprises 50% to 95% shielding particles and 50% to 5% silicon dioxide (SiO_2) filler particles.

* * * * *